higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

27 November 2025

This marking guideline consists of 8 pages.

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NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
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QUESTION 1: AC THEORY

1.1 1.1.1 It is a range of frequencies between f_H and f_L where the power has fallen.

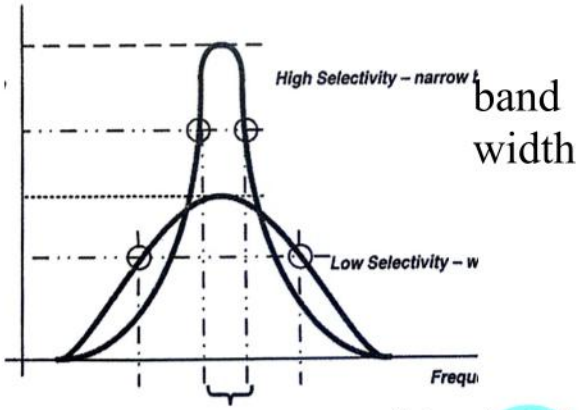
1.1.2 It is a figure of merit used to rate resonant circuits.

OR

It is the ratio of reactance of the inductor/capacitor to its equivalent resistance.

(2 × 2) (4)

1.2



(Allocate ONE mark for a correct, labelled drawing) (6)

1.3 1.3.1 $$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$L = \frac{1}{(2\pi f)^2 C} \checkmark$$

$$L = \frac{1}{(2\pi \times 350\,x\,10^3)^2 \times 56\,x\,10^{-12}}$$

$$L = 3{,}692\ \ mH$$

(4)

1.3.2 $$X_L = 2\pi fL$$

$$= 2\pi \times 350\,x\,10^3 \times 3{,}692\,x\,10^{-3}$$

$$= 8{,}12\ \ k\Omega \checkmark$$

(1)

[15]

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QUESTION 2: POWER SUPPLY

2.1
- Simple capacitor filter
- Series inductor filter
- LC-input filter
- RC-π-filter
- LC-π-filter

(Any 4 × 1) (4)

2.2 2.2.1

$$V'_{DC} = V_{DC} - I_{DC}R$$

$$= 150 - \left(100\,x\,10^{-3} \times 300\right)$$

$$= 120\ V \checkmark$$

2.2.2

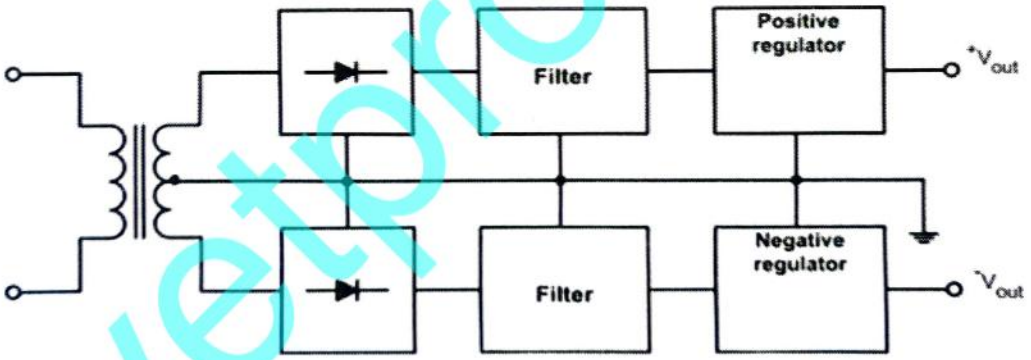
$$V'_{DC} = \frac{R_L}{R_L + R_S} V_{DC}$$

$$= \frac{1\ \ 200}{1\ \ 200 + 300} \times 150$$

$$= 120\ V \checkmark$$

(2 × 2) (4)

2.3



(Allocate ONE mark for a correct, labelled block) (7)

[15]

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QUESTION 3: AMPLIFIERS

3.1 $$V_E = \frac{V_{CC}}{10} = \frac{34}{10} = 3.4 \ V$$

$$I_E = I_C = \frac{V_E}{R_E} = \frac{3,4}{2000} = 1,7 \ mA$$

$$I_B = \frac{I_C}{\beta} = \frac{1,7 \times 10^{-3}}{200} = 8,5 \ \mu A$$

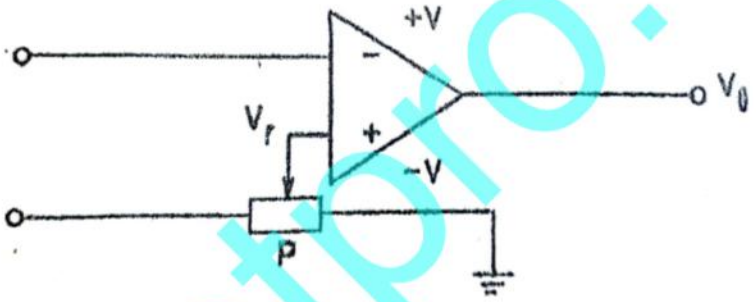
$$V_{CE} = V_{CC} - V_{RC} - V_E$$

$$= 34 - (1,7 \times 10^{-3} \times 5000) - 3,4$$

$$= 22,1 \ V$$ (9)

3.2 CMRR is the differential amplifier that has a high gain for differential mode signals and low gain for common mode signals.✓ (1)

3.3 $V_0 = A (V_r - V_i)$



(Allocate ONE mark for an amplifier and half a mark each for correctly labelled pins) (3)

3.4 $$f_O = \frac{1}{2\pi CR}$$

$$= \frac{1}{2\pi \times 20 \times 10^{-9} \times 120 \times 10^{3}}$$

$$= 66,314 \ Hz$$✓

(2)
[15]

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QUESTION 4: INTEGRATED CIRCUITS AND TRANSDUCERS

4.1
- Power amplification
- Small signal amplifiers
- Operational amplifiers
- Micro-wave amplifiers
- Voltage comparator
- Voltage regulator

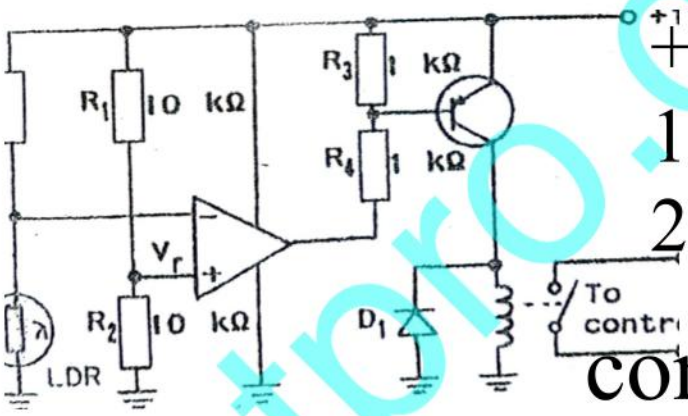
(Any 4 × 1) (4)

4.2 ESD is when two electrically charged objects, such as the human body and a electronic device meet and static electricity is discharged. (2)

4.3
- Ratio of transducer
- The fundamental transducer parameter
- Physical conditions
- Environmental conditions
- The adaptability with associated equipment

(Any 3 × 1) (3)

4.4



(Allocate half marks for correctly labelled electrical components) (5)

4.5
- AND-function
- OR-function

(Any 1 × 1) (1)

[15]

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QUESTION 5: ELECTRONIC PHASE CONTROL

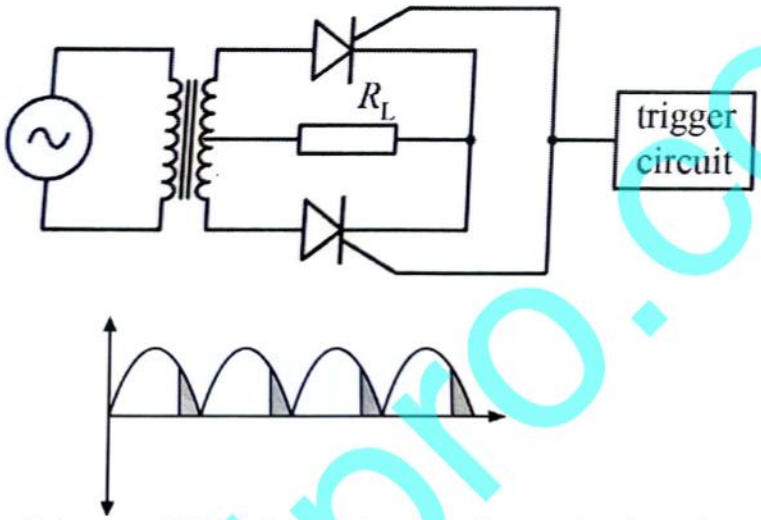
5.1 Delay angle – when SCR is ON but not conducting.✓

Conduction angle – when an SRC is ON and the gate is triggered allowing the current to pass through.✓ (2)

5.2
- Accuracy
- Sensitivity
- Noise
- Stability
- Bandwidth
- Speed
- Oscillation

(Any 4 × 1) (4)

5.3



(Allocate ONE full mark for the trigger circuit and output waveform, and a half mark for correct electric components) (4)
[10]

QUESTION 6: MEASURING INSTRUMENT

6.1
6.1.1 1000
6.1.2 12 V
6.1.3 Reset 1 to 0
6.1.4 Vr >Vu
6.1.5 1011

(5 × 1) (5)

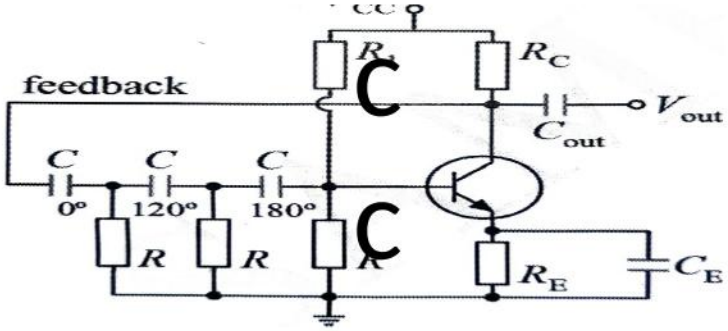
6.2 6.2.1 Convert analogue signals✓ to digital signals.✓ (2)

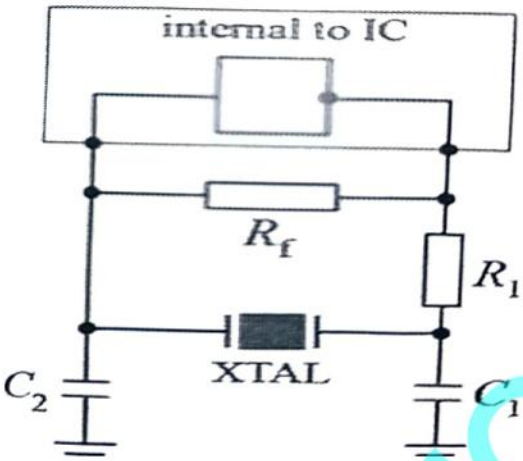
6.2.2 Will convert an incoming✓ analogue signal✓ to a discreet number of steps.✓ (3)
[10]

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QUESTION 7: MEASURING INSTRUMENT

7.1





(Allocate ONE mark for an RC-phase shift circuit, and ONE mark for a crystal controlled circuit) (2)

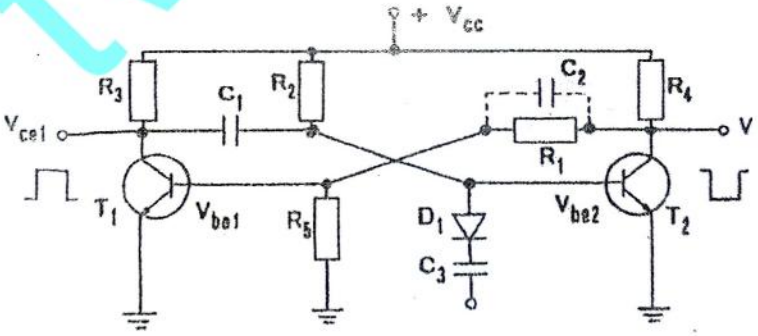
7.2

$$f_o = \frac{1}{1{,}4\,RC}$$

$$C = \frac{1}{1{,}4 \times 10000 \times 1{,}519}\checkmark\checkmark$$

$$= 47\ \mu F$$ (3)

7.3



(Allocate half marks for correctly labelled electrical components) (5)
[10]

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QUESTION 8

8.1
- Must use polarised light.✓✓
- The basic structure of the LCD should be controlled✓ by changing the applied current.✓
- The liquid crystal should control both operations to transmit✓ or can also change polarised light.✓ (6)

8.2 8.2.1 In a transmission cell both glass sheets are transparent so that the light is scattered in the forward direction✓ when the cell becomes active.✓

8.2.2 A reflective type cell consist of the reflecting surface of the glass sheet on one end.

(2 × 2) (4)
[10]

TOTAL: 100

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